INTERIM FEDERAL SPECIFICATION

PACKING MATERIAL (ASBESTOS, LUBRICATED)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D. C. 20407 based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. This specification covers braided, square lubricated asbestos packing used in pumps, valves and other equipment handling petroleum derivatives.

1.2 Classification. This specification covers only one type and grade of braided, square lubricated asbestos packing and shall be furnished in the sizes as specified (see 3.3 and 6.2).

2. APPLICABLE SPECIFICATIONS AND STANDARDS.

2.1 Specifications and Standards. The following specifications and standards of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard. Shipping
PPP-B-640 - Boxes, Folding, Fiberboard, Corrugated, Triple Wall.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Set-Up, Paperboard.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper (For Carton Sealing).

Federal Standards:

~~Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels.~~
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 601 - Rubber: Sampling and Testing.

(FOR INSTRUCTIONS ON OBTAINING COPIES OF FEDERAL SPECIFICATIONS AND STANDARDS, SEE LAST PAGE.)

Military Specifications:

MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-B-43014 - Boxes: Water Resistant Paperboard; Folding, Set-Up, and Metal-Stayed.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 Materials. The packing material shall be made of white asbestos yarn, lubricated with a compound unaffected by action of oils, fuels, solvents or other petroleum derivatives or products.

3.2 Construction.

3.2.1 Braid. Packing shall be either "braid over braid" or "regular braided", without wire insertion.

3.2.2 Cross-section. The packing shall be approximately square in cross-section. It shall be pressed or molded to conform to the size required (see table I).

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3.3 Size and weight. The sizes and weights shall be in accordance with table I.

Table I - Size and Weight

Nominal Size 1/	Weight, Per 100 Feet, Nominal
inch	pounds
1/8 2/	1.0
3/16 2/	2.5
1/4 2/	4.0
5/16 2/	5.75
3/8 2/	10.0
7/16 2/	11.5
1/2 2/	16.5
9/16 2/	20.0
5/8 2/	24.0
11/16	30.0
3/4	35.0
13/16	38.0
7/8	42.0
15/16	50.0
1	54.0

1/ All sizes available in 25, 50 and 100 pound reels.
2/ Available in 1/4, 1 or 5 pound spools or coils.

3.4 Yarn. The yarn shall have an asbestos content of not less than 80 per cent by weight (see 4.3.1).

3.5 Lubricant. Lubricating compound, exclusive of graphite shall not exceed 20 per cent, by weight of the finished packing (see 4.3.2).

3.6 Graphite. The finished packing shall be dipped in flake graphite.

3.7 Workmanship. Packing shall be free of voids, lumps, foreign particles or other defects that may impair its serviceability.

4. SAMPLING, INSPECTION AND TEST PROCEDURES.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all packing on spools, coils or reels of the same size, submitted for inspection at the same time.

4.2.2 Dimensional and visual examination of end item. Sampling for these characteristics shall be in accordance with level S-3 in MIL-STD-105.

4.2.3 Sampling for tests. Samples for all other tests shall be as follows:

Sample Size	Level or Number
2-15	0
16-50	S-1
51 and above	S-2

4.3 Inspection.

4.3.1 End item. Packing shall be examined for defects listed in table I. The Acceptable Quality Levels (AQL's) for defects listed in table I, shall be 2.5 per cent defective for major defects and 4.0 per cent defective for minor defects.

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Table II - Classification of Defects

Examine	Defects	Major	Minor
End Item	Dimension not as specified.	X	
	Voids or depressions in packing.	X	
	Lumps in packing.	X	
	Damaged or defective not affecting function or serviceability		X

4.3.2 Inspection of preparation for delivery requirements. An inspection shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with Table III. For examination of contents the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II and the AQL shall be 6.0 defects per hundred units.

Table III - Classification of Preparation for Delivery Defects

Examine	
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

4.4 Tests.

4.4.1 Asbestos. The asbestos content of the yarns shall be determined as described in method 16651 of Fed. Std. No. 601 (see 3.4).

4.4.2 Lubricant. The amount of organic solvent-soluable lubricant and the amount of graphite in the packing shall be determined as described in method 16627 of Fed. Std. No. 601 (see 3.4).

4.5 Rejection and retests. In case of failure of the test specimens to meet the requirements, samples may be resubmitted for test provided the producer has reworked the lots, as necessary to correct the deficiencies or has removed the nonconforming material.

5. PREPARATION FOR DELIVERY.

(For civil agency procurement the definitions and applications of levels of packaging and packing shall be in accordance with Fed. Std. No. 102).

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A. The packing furnished on spools, coils or reels in sizes and lengths as specified (see 6.2), shall be wrapped with material conforming to MIL-B-121, grade A, class 2 and packaged in a close-fitting box conforming to MIL-B-43014 or PPP-B-636, class 2, RSC, as applicable. The box shall be closed in accordance with the appendix to the applicable box specification. WEATHER RESISTANT

5.1.2 Level B. The packing furnished on spools, coils or reels in sizes and lengths as specified (see 6.2), shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665, PPP-B-676 or PPP-B-636, class 1, style RSC, as applicable. The box shall be closed with 2-inch wide tape conforming to PPP-T-76 on the top and bottom seams. DOMESTIC

5.1.3 Level C. The packing shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Packing of like description packaged as specified in 5.1.1 shall be packed together in close-fitting boxes conforming to PPP-B-585, class 3, PPP-B-591, class 2 PPP-B-601, type II, class 2, PPP-B-621, class 2, PPP-B-636, class 2 or PPP-B-640, class 2, grade A. The fiberboard boxes shall be closed

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in accordance with the appendix to the applicable box specification. The gross weight of the triple wall fiberboard box and the wood boxes shall not exceed approximately 200 pounds. The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification. Strapping shall not exceed the weight limitations of the box specification. Strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Packing of like description packaged as specified in 5.1.2 shall be packed together in close-fitting boxes conforming to PPP-B-585, class 1, PPP-B-591, class 1, PPP-B-601, domestic type, PPP-B-621, class 1 and PPP-B-636, class 1. The fiberboard box shall be closed in accordance with the appendix to the box specification. The gross weight of the wood boxes shall not exceed approximately 200 pounds and the fiberboard boxes shall not exceed the wieght limitations of the box specification.

5.2.3 Level C. The packing shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. This specification covers packing for use on centrifugal and rotary pumps, turning shafts, valve stems, fluid end rods on equipment handling gasoline, solvents and other thin and agressive petroleum derivatives that would wash the lubricant out of general service packings. Speeds up to 3600 RPM; temperatures to 500°F.

6.2 Ordering data, Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size and weight of coil required (see 3.3).
(c) Size of lot, if different from 4.2.1.
(d) Level of packaging and packing required (see 5.1 and 5.2).

6.3 Asbestos packing in accordance with this specification is usually purchased by weight.

6.4 Transportation description. The transportation descriptions and minimum weights applicable to this commodity are:

RAIL:

MOTOR:

Military Interests
Army - MR
Navy - SHIPS
Air Force - MOA

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